DATA CLEANING DOCUMENTATION FOR WHO TB BURDEN ANALYSIS

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1. Purpose

This document records how the WHO TB data was prepared for analysis. It lists the columns kept, how they were renamed and defined, and how blank values and data types were handled, and the checks that were run, so the cleaning can be checked and repeated.

2. Data Source

WHO Global Tuberculosis Programme data (WHO Global TB Report 2025, with estimates up to 2024). The files were downloaded on 16 October 2026 from

https://www.who.int/teams/global-tuberculosis-programme/data

File downloaded	Contents	Rows in source
TB_burden_countries_2026-10-16.csv	Burden estimates: cases, deaths, case detection	5,347
TB_outcomes_2026-10-16.csv	Treatment outcomes	6,399
MDR_RR_TB_burden_estimates_2026-10-16.csv	Drug-resistant TB estimates	2,150
TB_data_dictionary_2026-10-16.csv	Definitions of every WHO column	not used for analysis

The burden file has 50 columns and the outcomes file has 73, so only the columns needed to answer the analysis questions were kept. The three data files share a country code (iso3) and year, which are the join keys.

3. Tools Used

- Excel Power Query: selecting columns, renaming, setting data types, exporting clean CSV files.
- MySQL (Workbench): importing the clean files and running checks.

4. Cleaning steps

Step 1: Select the columns needed

Using Power Query's "Remove other columns", only the columns below were kept.

Table	Columns kept
tb_burden_estimates	country, iso3, g_whoregion, year, e_pop_num, e_inc_100k, e_inc_num, e_mort_100k, e_mort_num, c_cdr
treatment_outcomes	iso3, year, newrel_coh, newrel_succ, c_new_tsr, mdr_coh, mdr_succ
mdr_rr_burden_estimates	iso3, year, e_rr_pct_new, e_rr_pct_ret, e_inc_rr_num

Step 2: Rename columns

WHO's column names are coded, so they were renamed in lowercase with underscores. Definitions come from the WHO data dictionary.

S/N	Table	Original lumn names	Renamed columns	Definition
1	Tb_burden_estimates	country	country	Country me
2	Tb_burden_estimates	iso3	iso3	3-letter untry code
3	Tb_burden_estimates	year	year	Year
4	Tb_burden_estimates	g_whoregion	who_region	WHO gion
5	Tb_burden_estimates	e_pop_num	population	Estimated tal pulation
6	Tb_burden_estimates	e_inc_100k	incidence_per_100k	Estimated w cases (all rms) per 0,000 ople
7	Tb_burden_estimates	e_inc_num	incident_cases	Estimated mber of new B cases (all rms) that ar
8	Tb_burden_estimates	e_mort_100k	deaths_per_100k	Estimated B deaths per 0,000 ople
9	Tb_burden_estimates	e_mort_num	deaths	Estimated tal number TB deaths
10	Tb_burden_estimates	c_cdr	case_detection_pct	Percentage estimated ses that were und and ported (also own as TB eatment verage)
11	Mdr_rr_burden_estimates	iso3	iso3	3-letter untry code
12	Mdr_rr_burden_estimates	year	year	Year

13	Mdr_rr_burden_estimates	e_rr_pct_new	rr_pct_new	Estimated rcentage of w TB cases at are fampicin-sistant.
14	Mdr_rr_burden_estimates	e_rr_pct_ret	rr_pct_retreated	Estimated rcentage of eviously eated TB ses that are fampicin-sistant.
15	Mdr_rr_burden_estimates	e_inc_rr_num	rr_cases_est	Estimated mber of fampicin-sistant cases ew and eviously eated).
16	Treatment_outcomes	iso3	iso3	3-letter untry code
17	Treatment_outcomes	year	year	year
18	Treatment_outcomes	newrel_coh	new_cohort	Number of w and lapse tients who arted eatment that ar.
19	Treatment_outcomes	newrel_succ	new_success	Number of ose patients ho were red or mpleted eatment
20	Treatment_outcomes	c_new_tsr	new_succ_rate_pct	Percentage the cohort hose eatment was ccessful.
21	Treatment_outcomes	mdr_coh	mdr_cohort	Number of DR-TB tients who arted eatment.

22	Treatment_outcomes	mdr_succ	mdr_success	Number of DR-TB tients who ere cured or mpleted eatment

Step 3: Set data types

In Power Query, year and counts were set to whole numbers, and percentages and rates to decimal numbers, so that no decimals were lost.

Step 4: Handle blank values

Blank cells were kept as blanks (they become NULL in MySQL). No rows were removed in Power Query, which keeps the data as WHO published it. Row counts after loading matched the source files: 5,347, 2,150 and 6,399.

Step 5: Export and import into MySQL

The cleaned tables were exported as CSV files and imported with MySQL Workbench's Table Data Import Wizard. The wizard drops rows that have a blank in a numeric column, so columns with blanks were imported as TEXT first. Blanks were then converted to NULL, and the columns were changed to their numeric types:

1. Imported the file with the blank-heavy columns set to TEXT.
2. Ran UPDATE(…) to turn empty text into NULL for each column.
3. Ran ALTER TABLE ... MODIFY to change the columns to INT, BIGINT or DOUBLE.

Type	**Columns**
VARCHAR(3)	iso3
VARCHAR(100)	country
VARCHAR(10)	who_region
INT	year, incident_cases, deaths, rr_cases_est, new_cohort, new_success, mdr_cohort, mdr_success
BIGINT	population
DOUBLE	incidence_per_100k, deaths_per_100k, case_detection_pct, rr_pct_new, rr_pct_retreated, new_succ_rate_pct

Step 6: Data quality checks

Check	**Result**
Row count, tb_burden_estimates	5,322 (source 5,347; the 25 missing rows are DPRK, see section 6)
Row count, mdr_rr_burden_estimates	2,150, matches source
Row count, treatment_outcomes	6,399, matches source

Check	Result
Nigeria rows in tb_burden_estimates	25 (2000 to 2024)
NULLs in case_detection_pct	339
Blank rows in mdr_rr_burden_estimates	330 (blank in all 3 estimate columns)
Blanks in treatment_outcomes	new_cohort 4,058; new_success 4,070; new_succ_rate_pct 1,102; mdr_cohort 3,355; mdr_success 3,438

Step 7: Issues found and decisions

Issue	Decision
MySQL import wizard dropped rows with blank numeric cells	Imported blank-heavy columns as TEXT, converted blanks to NULL, then changed to numeric types
DPRK has blank estimates for every year, so its 25 rows were not loaded	Accepted: the rows contained no data. Noted as a limitation.
330 resistance-table rows are blank (countries and territories with no estimates)	Kept as NULL; use LEFT JOIN so these countries are not lost
WHO's rounded percentages create ties when ranking	Rates are recalculated from the raw counts (for example new_success divided by new_cohort)
Country names can differ between files, and "Congo" and "Democratic Republic of the Congo" are different countries	Tables are joined on iso3 (and year), not on country name
MDR treatment outcomes end at 2022 (2023 is blank)	Each indicator uses the latest year available, shown in every chart title

Step 8: Final clean tables

Table	Rows	Used for
tb_burden_estimates	5,322	Cases, rates, deaths, case detection
treatment_outcomes	6,399	Treatment success
mdr_rr_burden_estimates	2,150	Drug resistance
high_burden_30	30	The 30 high-burden countries (WHO method applied to 2024 data)

Step 9: Limitations

- WHO figures are estimates from statistical models and may be revised in later reports.
- Several columns have many blanks, mostly for earlier years and small countries, so comparisons are limited to countries with data.
- Only the columns needed for this analysis were kept; the other WHO columns (for example uncertainty ranges and age groups) were not used.